```
Dataset shape: torch.Size([200, 10])
Labels shape: torch.Size([200])

First 5 sequences:
tensor([[ 3,  4, 12, 17,  8, 18, 16,  8,  4,  8],
        [16,  0,  0,  1, 18, 17, 18, 10, 10, 14],
        [ 7, 12,  4,  4, 12, 11, 10, 12, 18, 18],
        [ 7,  0,  7, 12,  6, 11,  5, 19, 16,  8],
        [12, 15, 19,  6,  5, 19,  2,  5, 17, 19]])

First 5 labels:
tensor([0, 0, 0, 1, 1])

Training data: torch.Size([150, 10])
Testing data: torch.Size([50, 10])
Training labels: torch.Size([150])
```

```
TinyTransformer(
  (emb): Embedding(20, 32)
  (trans): TransformerEncoder(
    (layers): ModuleList(
      (0-1): 2 x TransformerEncoderLayer(
        (self_attn): MultiheadAttention(
          (out_proj): NonDynamicallyQuantizableLinear(in_features=32, out_features=32, bias=True)
        )
        (linear1): Linear(in_features=32, out_features=2048, bias=True)
        (dropout): Dropout(p=0.1, inplace=False)
        (linear2): Linear(in_features=2048, out_features=32, bias=True)
        (norm1): LayerNorm((32,), eps=1e-05, elementwise_affine=True)
        (norm2): LayerNorm((32,), eps=1e-05, elementwise_affine=True)
        (dropout1): Dropout(p=0.1, inplace=False)
        (dropout2): Dropout(p=0.1, inplace=False)
      )
    )
  )
  (fc): Linear(in_features=32, out_features=2, bias=True)
)

Model created successfully!
```

```
Epoch 1 Loss 0.6930498480796814
Epoch 2 Loss 0.7004915475845337
Epoch 3 Loss 0.6978026628494263
Epoch 4 Loss 0.6908214688301086
Epoch 5 Loss 0.6976053714752197
Epoch 6 Loss 0.6996820569038391
Epoch 7 Loss 0.6968796849250793
Epoch 8 Loss 0.6973799467086792
Epoch 9 Loss 0.6974911689758301
Epoch 10 Loss 0.6961090564727783
```

```
Actual labels:     tensor([0, 0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 1, 1, 1, 0, 0,
        0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1, 0, 0, 0, 1,
        1, 1])
Predicted labels:  tensor([1, 1, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 1, 1, 1,
        0, 1, 0, 1, 1, 1, 1, 1, 0, 0, 1, 1, 1, 1, 1, 0, 0, 1, 1, 1, 1, 1, 1, 1,
        1, 0])
Test accuracy: 0.5199999809265137
Test accuracy (%): 51.99999809265137
```